PBH Formation from QCD-like Equation of State Transitions with Numerical Relativity

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1 Abstract

We study primordial black hole (PBH) formation during QCD-like equation-of-state transitions using relativistic simulations of spherically symmetric collapse in an expanding cosmological background. Unlike constant-w treatments, time-dependent crossovers generally produce $c_s^2 \neq w$, separating the effects of background expansion from pressure-wave propagation. We investigate how the depth and duration of QCD-like softening modify the critical collapse threshold δ_c. The numerical solver reproduces the standard radiation-era threshold agreeing within 1.3% relative to the analytic expectations. For transitions with $w_{\min} = 0.12$, the collapse threshold decreases by up to 44% for larger crossovers, implying exponentially enhanced PBH production in Press–Schechter estimates. The enhancement peaks near $T \simeq 132$ MeV, corresponding to PBHs of approximately solar mass. These results show that the time structure of QCD-era softening can significantly affect PBH formation and motivate more complete lattice-QCD and multidimensional numerical-relativity studies.

2 Introduction

Primordial black holes are hypothetical black holes that may have formed in the early universe before the existence of stars, galaxies, or other astrophysical structures. Unlike stellar-mass black holes, which arise from the collapse of massive stars, primordial black holes could form directly from large density perturbations in the early universe. If a region of the universe was sufficiently overdense when it entered the cosmological horizon, its self-gravity could overcome pressure support and cosmic expansion, causing the region to collapse into a black hole [1, 2, 3]. This makes primordial black holes an important theoretical probe of both cosmology and high-energy physics. PBHs remain an active subject of study because they could contribute to dark matter, seed structure formation, or leave observable gravitational-wave signatures, although current observational constraints rule out PBHs as all of the dark matter over many mass ranges [4].

The central quantity controlling PBH formation is the critical density threshold, usually denoted δ_c. A density perturbation with amplitude $\delta < \delta_c$ disperses or is diluted by cosmic expansion, while a perturbation with $\delta > \delta_c$ can collapse to form a black hole. The value of δ_c is therefore not merely a technical detail. PBH abundance is exponentially sensitive to this threshold. In Press–Schechter-type estimates, the formation fraction behaves approximately as

$$\beta \sim \exp\left(-\frac{\delta_c^2}{2\sigma_\delta^2}\right), \tag{1}$$

where σ_δ is the standard deviation of the primordial density perturbations on the relevant scale. Because δ_c appears in the exponent, even a relatively small reduction in the collapse threshold can lead to orders-of-magnitude enhancement in the number of PBHs formed.

Early estimates by Carr and Hawking related the collapse threshold to the equation-of-state parameter

$$w = \frac{p}{\rho}, \tag{2}$$

where p is pressure and ρ is energy density [2, 3]. In a radiation-dominated universe, $w = 1/3$, and pressure gradients resist collapse. Softer equations of state, with smaller w, reduce pressure support and therefore make gravitational collapse easier. Modern calculations show that the threshold also depends on the shape of the perturbation profile, the sound speed, the slicing used to define the overdensity, and relativistic effects near horizon entry [7, 9, 10].

This observation is especially important during the QCD crossover. At temperatures of order $T \sim 150$ MeV, the universe transitioned from a quark-gluon plasma to a hadron-dominated state. In the standard model at low baryon chemical potential, this transition is not expected to be a sharp first-order phase transition, but rather a smooth crossover [11]. During this crossover, the effective equation of state softens. The parameter w can drop below its radiation value of $1/3$, and the sound speed can also decrease. This temporary reduction in pressure support makes the QCD epoch a natural period during which PBH formation may be enhanced. Recent lattice-QCD work continues to refine the thermodynamics of the QCD crossover at low temperatures and finite baryon chemical potential [12].

The idea that QCD-era pressure softening enhances PBH formation is not new. Jedamzik argued that the QCD epoch could produce a pronounced solar-mass PBH feature because of the reduction in effective sound speed [6]. Semi-analytic work by Sobrinho, Augusto, and Goncalves studied threshold reductions during QCD-era models, including bag-model, lattice-fit, and crossover descriptions [13]. Byrnes, Hindmarsh, Young, and Hawkins used lattice-QCD thermodynamics to estimate PBH mass-function enhancement and found a peak near the solar-mass scale [14]. Papanikolaou extended the Harada–Yoo–Kohri threshold argument toward a time-dependent equation-of-state background [15]. More recently, Escrivà, Bagui, and Clesse simulated PBH formation while taking the varying QCD equation of state into account, and Musco, Jedamzik, and Young presented a detailed numerical study of QCD-era PBH thresholds, mass distributions, and abundances [16, 17].

Therefore, this paper does not claim to be the first numerical treatment of PBH formation during the QCD crossover. The narrower purpose of this work is to isolate, in a controlled QCD-like model, how the duration and depth of a time-dependent equation-of-state softening affect the collapse threshold. This is a sensitivity study rather than a full lattice-QCD calculation. We evolve spherically symmetric overdensities on an expanding FLRW background with time-dependent equation-of-state models. We compare constant radiation, Gaussian QCD-like crossover, parametric crossover, and lattice-QCD-inspired equations of state. The collapse dynamics are evolved using a Misner–Sharp relativistic formulation, and the results are checked against the analytic threshold formula in the radiation-dominated limit. We then use the simulation and semi-analytic threshold estimates to determine how QCD-like softening changes the critical threshold and the resulting PBH abundance.

A subtle but important point is that, when the equation of state varies in time, the thermodynamic sound speed is not generally equal to w. For a barotropic constant equation of state, one often has $c_s^2 = w$. However, if $p = w(t)\rho$ and the background obeys energy conservation, then

$$c_s^2 = \frac{\dot{p}}{\dot{\rho}} = w - \frac{\dot{w}}{3H(1+w)}. \tag{3}$$

This distinction matters because pressure waves are responsible for resisting collapse. The background expansion depends on $w(t)$, but the propagation of pressure support depends on $c_s^2(t)$. With the sign convention above and $H > 0$, a decreasing w gives $c_s^2 > w$, while an increasing w gives $c_s^2 < w$. Thus the key point is not simply that the entry phase of the crossover suppresses the sound speed. Rather, the time dependence of the crossover can either raise or lower c_s^2 relative to w depending on the sign of $\dot{w}$, and a constant-w approximation cannot capture that asymmetry.

The main result is that QCD-like softening can substantially lower the collapse threshold in this model. Our numerical relativity calculation reproduces the standard radiation-era threshold with 1.3% agreement, giving $\delta_c^{\mathrm{NR}} = 0.419$ compared with an analytic estimate of 0.414. For QCD-like transitions with minimum equation-of-state parameter $w_{\min} = 0.12$, we find threshold reductions of 27.0% for a narrower crossover and 44.1% for a broader crossover. These reductions imply large Press–Schechter abundance enhancements for the chosen perturbation amplitude, reaching a peak enhancement of approximately 4.18×10^{10} for $\sigma_\delta = 0.03$. Because this enhancement is exponentially sensitive to δ_c and depends strongly on the assumed primordial perturbation variance, it should be interpreted as a model-dependent estimate, not a universal prediction of the QCD epoch.

3 Formulation and Numerical Method

3.1 Cosmological Background

We consider PBH formation in an expanding Friedmann–Lemaitre–Robertson–Walker background. The background dynamics are governed by the Friedmann equation

$$H^2 = \frac{8\pi G}{3}\rho, \tag{4}$$

together with energy conservation,

$$\dot{\rho} + 3H(\rho + p) = 0. \tag{5}$$

Writing the pressure as $p = w(t)\rho$, the Hubble parameter evolves according to

$$\dot{H} = -\frac{3}{2}(1 + w(t))H^2. \tag{6}$$

For constant radiation domination, $w = 1/3$ and $H = 1/(2t)$. In the QCD crossover case, however, w varies with time, so the simulation evolves $H(t)$ numerically rather than assuming the pure radiation-era solution.

3.2 Equation of State Models

The simulation includes several equation-of-state models. The constant equation-of-state model sets $w =$ constant and $c_s^2 = w$, with the radiation case given by $w = 1/3$. The QCD crossover model represents the softening of the equation of state with a Gaussian dip,

$$w(t) = \frac{1}{3} - \Delta w \exp\left[-\frac{(t - t_{\rm cross})^2}{2\sigma^2}\right], \tag{7}$$

where Δw controls the depth of the transition and σ controls its duration. The simulation also includes a parametric QCD-like model and a lattice-QCD-inspired equation of state based on effective degrees of freedom for energy and entropy.

For a time-dependent equation of state, the sound speed is not generally equal to w. Instead, the thermodynamic relation used in the simulation is

$$c_s^2 = w - \frac{\dot{w}}{3H(1+w)}. \tag{8}$$

For the Gaussian model,

$$\dot{w}(t) = \Delta w \, \frac{t - t_{\rm cross}}{\sigma^2} \exp\left[-\frac{(t - t_{\rm cross})^2}{2\sigma^2}\right]. \tag{9}$$

Thus $\dot{w} < 0$ before the minimum of the dip and $\dot{w} > 0$ after the minimum. The derivative term therefore changes sign across the crossover: the sound speed is raised relative to w on one side of the transition and suppressed relative to w on the other. Since pressure gradients resist collapse through sound-wave propagation, this difference between $w(t)$ and $c_s^2(t)$ is one of the key physical effects of the model.

3.3 Misner–Sharp Relativistic Collapse Solver

The collapse calculation uses a spherically symmetric relativistic formulation in comoving coordinates. The metric is written as

$$ds^2 = -A^2 dt^2 + B^2 dr^2 + R^2 d\Omega^2, \tag{10}$$

where $A(t,r)$ is the lapse function, $B(t,r)$ is the radial metric component, and $R(t,r)$ is the areal radius. The evolved variables are the areal radius $R(t,r)$, the proper radial velocity

$$U = D_t R, \tag{11}$$

and the Misner–Sharp mass $M(t,r)$, which represents the quasi-local gravitational mass inside radius R. The main evolution equations are

$$\frac{dR}{dt} = AU, \tag{12}$$

$$\frac{dU}{dt} = -A\frac{\Gamma}{B(\rho+p)}\frac{\partial p}{\partial r} - A\frac{M + 4\pi R^3 p}{R^2}, \tag{13}$$

$$\frac{M}{dt} = -4\pi R^2 p A U. \tag{14}$$

The quantity Γ is defined by

$$\Gamma^2 = 1 + U^2 - \frac{2M}{R}. \tag{15}$$

A trapped surface corresponds to the limit where $2M/R$ becomes large enough that Γ approaches zero. This makes Γ a useful collapse diagnostic.

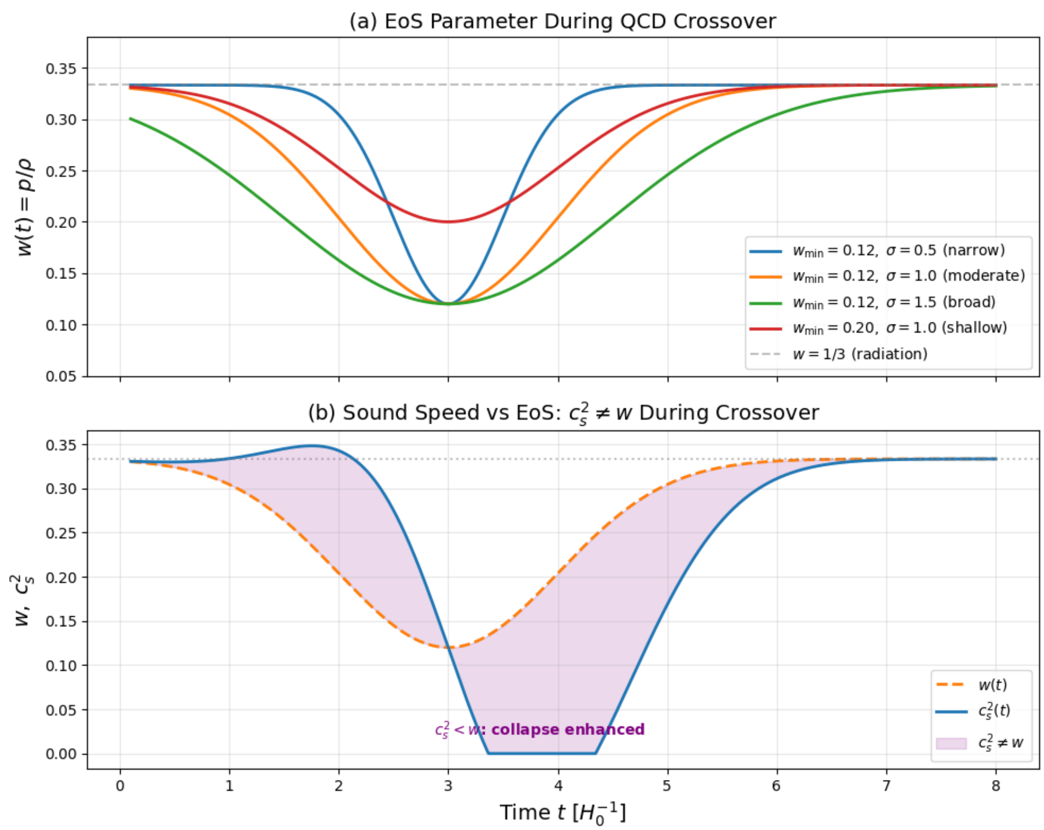

Figure 1: (a) Evolution of the equation-of-state parameter $w(t)$ and (b) sound speed $c_s^2(t)$ during the QCD crossover. The top panel illustrates the softening of $w(t)$ across narrow, moderate, and broad transition widths. The bottom panel demonstrates the resulting time-asymmetry in the sound speed c_s^2, highlighting the regime where $c_s^2 < w$, which enhances the probability of collapse.

The mass constraint is

$$\frac{\partial M}{\partial r} = 4\pi\rho R^2 \frac{\partial R}{\partial r}, \tag{16}$$

and the lapse is determined from the momentum constraint,

$$\frac{\partial \ln A}{\partial r} = -\frac{\partial p/\partial r}{\rho + p}. \tag{17}$$

Together, these equations evolve the overdense region while preserving the relativistic relation between matter, geometry, and expansion.

3.4 Initial Conditions

The initial density perturbation is taken to be a Gaussian overdensity on top of the FLRW background:

$$\rho(r) = \rho_{\mathrm{bg}} \left[1 + \delta_0 \exp\left(-\frac{r^2}{2r_p^2}\right)\right], \tag{18}$$

where δ_0 is the initial amplitude and r_p is the characteristic perturbation width. The initial velocity is chosen to match the background Hubble flow,

$$U(r) = H_0 R. \tag{19}$$

Thus the overdense region initially participates in cosmic expansion, but the excess mass creates additional gravitational binding. The Misner–Sharp mass is then obtained by integrating the mass constraint.

3.5 Time Integration and Collapse Criteria

The equations are evolved using fourth-order Runge–Kutta integration with adaptive CFL time stepping. The timestep is chosen according to the maximum characteristic propagation speed, including both fluid motion and sound speed. Artificial viscosity is included to stabilize the evolution in regions where sharp gradients or shocks may develop. Collapse is diagnosed using two independent criteria. First, the compaction function

$$C = \frac{2\delta M}{R} \tag{20}$$

is monitored, with collapse identified when $C \gtrsim 0.42$. Second, the simulation tracks the minimum value of Γ, with $\Gamma_{\min} < 0.10$ indicating approach toward trapped-surface formation. Agreement between these diagnostics provides a consistency check on the collapse threshold.

3.6 Semi-Analytic Threshold and Abundance Estimate

To compare with semi-analytic expectations, we use a generalized threshold expression motivated by the Harada–Yoo–Kohri formula [9, 15]. In the time-dependent case, the equation of state and sound speed are averaged over the relevant sound-crossing timescale:

$$\delta_c = \frac{3(1 + w_{\mathrm{eff}})}{5 + 3w_{\mathrm{eff}}} \sin^2\left[\frac{\pi\sqrt{c_{s,\mathrm{eff}}^2}}{1 + 3w_{\mathrm{eff}}}\right]. \tag{21}$$

The prefactor depends on the background equation of state, while the sine-squared term depends on the sound speed because it describes pressure-wave propagation across the overdense region. The PBH abundance is estimated using a Press–Schechter-type expression,

$$\beta \sim \exp\left(-\frac{\delta_c^2}{2\sigma_\delta^2}\right). \tag{22}$$

The enhancement relative to pure radiation domination is therefore

$$\frac{\beta_{\rm QCD}}{\beta_{\rm rad}} = \exp\left[\frac{\delta_{c,\rm rad}^2 - \delta_{c,\rm QCD}^2}{2\sigma_\delta^2}\right]. \tag{23}$$

This exponential sensitivity is why even moderate reductions in δ_c can produce very large changes in the predicted PBH abundance. It is also why abundance comparisons between different papers must be made cautiously: different perturbation spectra, threshold definitions, equation-of-state models, and values of σ_δ can lead to very different enhancement factors.

4 Results and Interpretation

The simulation was first tested in the constant radiation-dominated case, where $w = 1/3$ and $c_s^2 = w$. This case provides an important validation benchmark because it corresponds to the standard PBH formation scenario studied in much of the literature [7, 8, 9, 10]. For the chosen perturbation profile, the numerical relativity solver gives

$$\delta_c^{\rm NR} = 0.419. \tag{24}$$

The corresponding analytic estimate gives

$$\delta_c^{\rm analytic} = 0.414. \tag{25}$$

The fractional difference is approximately 1.3%. This agreement is important because it shows that the simulation correctly captures the balance between pressure support, gravity, and cosmic expansion in the known radiation-era limit. The QCD crossover results are therefore not simply artifacts of the numerical method; they are obtained using a solver that reproduces the expected threshold in the constant-w case.

After this validation, we introduced a QCD-like time-dependent equation of state. The main physical effect of the crossover is a temporary reduction in pressure support. In a pure radiation background, relativistic pressure gradients resist collapse efficiently. During the QCD crossover, however, the equation of state softens, lowering w below $1/3$. In addition, because w changes with time, the sound speed no longer equals w. The sound speed is affected by $\dot{w}$, which changes the ability of pressure waves to cross the overdense region before collapse occurs. This distinction between w and c_s^2 is one of the central physical effects in the simulation. The background expansion rate is controlled by the equation-of-state parameter $w(t)$, while pressure gradients depend on the sound speed $c_s^2(t)$. However, the sign of the derivative term matters. For the convention used here, $\dot{w} < 0$ gives $c_s^2 > w$, while $\dot{w} > 0$ gives $c_s^2 < w$. The result is a time-asymmetric crossover: pressure-wave propagation is not described by the instantaneous value of w alone. Collapse is controlled by how the overdensity horizon-entry time, sound-crossing time, and recovery of the equation of state overlap.

For a QCD-like Gaussian crossover with minimum equation-of-state parameter

$$w_{\rm min} = 0.12, \tag{26}$$

the simulation finds a significant reduction in the collapse threshold. For a crossover width $\sigma = 0.8$, the threshold is reduced by approximately

$$27.0\%. \tag{27}$$

For a broader crossover with $\sigma = 1.2$, the reduction increases to approximately

$$44.1\%. \tag{28}$$

The dependence on crossover width has a clear physical interpretation. A narrow crossover reduces pressure support only briefly. A broader crossover keeps the fluid in a softened state for a longer time, giving the overdense region more time to collapse before pressure gradients can halt the infall. Therefore, broader QCD-like transitions produce a larger reduction in δ_c in this model.

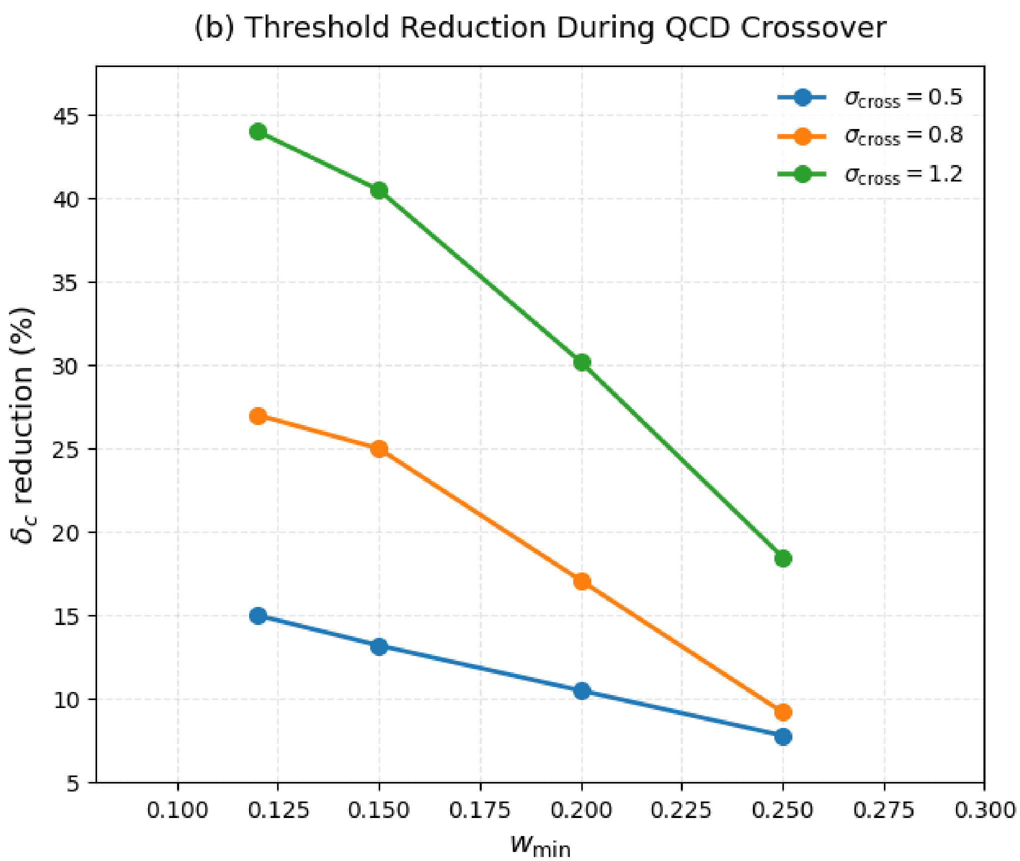

Figure 2: Threshold reduction during the QCD crossover. The percentage reduction in the critical collapse threshold (δ_c) is shown as a function of the minimum equation-of-state parameter (w_{min}). The different curves represent varying crossover durations (σ_{cross}), indicating that broader transitions yield substantially larger reductions in the collapse threshold.

The threshold reduction is not merely a small numerical correction. Since PBH abundance depends exponentially on δ_c^2, even a modest lowering of the collapse threshold can produce a very large increase in PBH formation. Using the Press–Schechter approximation,

$$\beta \sim \exp\left(-\frac{\delta_c^2}{2\sigma_\delta^2}\right), \tag{29}$$

the enhancement relative to the radiation-dominated case is

$$\frac{\beta_{\rm QCD}}{\beta_{\rm rad}} = \exp\left[\frac{\delta_{c,\rm rad}^2 - \delta_{c,\rm QCD}^2}{2\sigma_\delta^2}\right]. \tag{30}$$

For $\sigma_\delta = 0.03$, the peak enhancement in the simulation reaches approximately

$$4.18 \times 10^{10}. \tag{31}$$

This result illustrates the main cosmological implication of the work: the QCD crossover can convert a small change in the collapse threshold into a very large change in PBH abundance.

However, this number should not be compared directly to lattice-QCD-based abundance enhancements without matching the primordial power spectrum, the threshold convention, and the equation-of-state model. For example, Musco, Jedamzik, and Young found a QCD-scale enhancement of order 10^3 for nearly scale-invariant spectra in a detailed lattice-QCD-based numerical study [17]. Our larger enhancement follows from the larger threshold reductions in the chosen parametric model and the selected value $\sigma_\delta = 0.03$. The enhancement peaks near

$$T \simeq 132 \text{ MeV}. \tag{32}$$

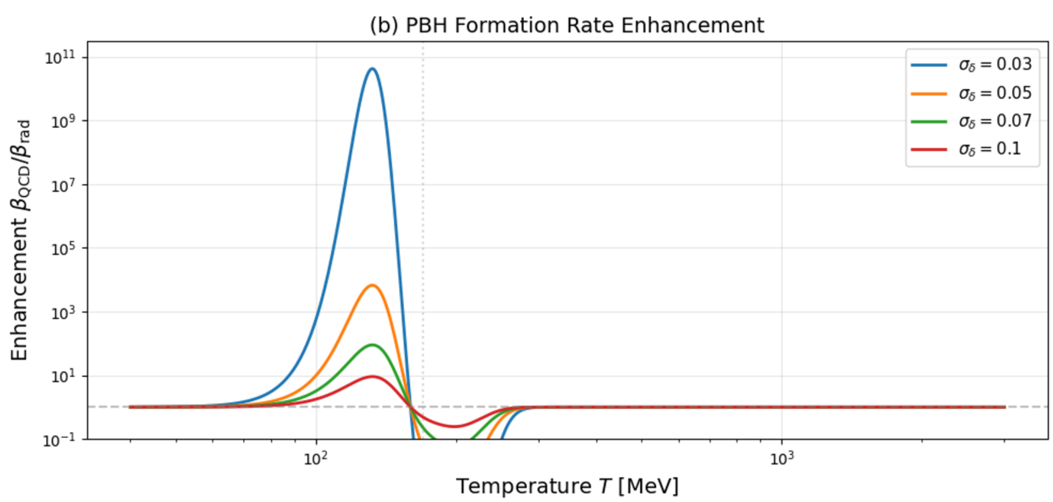

Figure 3: PBH formation rate enhancement factor ($\beta_{\rm QCD}/\beta_{\rm rad}$) plotted against cosmological temperature T [MeV] for different standard deviations of the primordial perturbation amplitude (σ_δ). The rate exhibits a pronounced peak near $T \simeq 132$ MeV, driven by the varying equation-of-state dynamics during the QCD crossover epoch.

This temperature is slightly below the nominal QCD crossover scale. The location of the peak is physically reasonable because the largest enhancement does not necessarily occur exactly at the point where w is minimized. Instead, it occurs when the time dependence of the equation of state, the sound-speed history, the horizon-entry time of the perturbation, and the collapse timescale combine most efficiently. In other words, PBH formation is controlled not only by the minimum value of w, but also by when the perturbation enters the horizon relative to the evolving crossover.

The characteristic PBH mass associated with the QCD epoch is of order

$$M_{\rm PBH} \sim 1 M_\odot. \tag{33}$$

This places QCD-era PBHs in a mass range relevant to stellar-mass compact objects and gravitational-wave observations. The simulation does not prove that such PBHs make up dark matter, nor does it predict a unique abundance without specifying the primordial perturbation spectrum. However, it does show that QCD-like equation-of-state softening can strongly amplify PBH production at approximately solar masses if sufficiently large primordial perturbations exist on the corresponding scale.

Quantity	Value	Interpretation
δ_c^{NR}	0.419	Radiation-era numerical threshold
$\delta_c^{\mathrm{analytic}}$	0.414	Analytic radiation-era benchmark
Validation error	1.3%	Confirms solver accuracy
Threshold reduction	27.0%	$w_{\mathrm{min}} = 0.12$, $\sigma = 0.8$
Threshold reduction	44.1%	$w_{\mathrm{min}} = 0.12$, $\sigma = 1.2$
Peak enhancement	4.18×10^{10}	$\sigma_\delta = 0.03$
Peak temperature	132 MeV	Near the QCD crossover
Characteristic mass	$\sim 1M_\odot$	Solar-mass PBH scale

Table 1: Summary of the main numerical and semi-analytic results.

The most important conclusion is that the QCD crossover affects PBH formation in two related but distinct ways. First, the decrease in w changes the background expansion and lowers the effective pressure of the cosmological fluid. Second, the time dependence of w changes the sound speed, which directly affects how pressure gradients propagate through the overdense region. The second effect is especially important because the sound speed controls whether pressure waves can prevent collapse before a trapped region begins to form. The result should be understood as an explicit parametric demonstration of this sensitivity, not as a claim that previous numerical QCD studies omitted the crossover.

There are also important limitations. The collapse calculation assumes spherical symmetry, whereas realistic primordial perturbations may be nonspherical. The initial perturbation profile is Gaussian, while the true primordial perturbation spectrum may contain a variety of shapes. The QCD crossover is prescribed through an equation-of-state model rather than evolved self-consistently from microscopic QCD degrees of freedom. Finally, the compaction-based threshold in the QCD case is measured relative to the evolving background, which means the semi-analytic threshold comparison remains important for interpreting the numerical result. Despite these limitations, the simulation provides evidence that a time-dependent QCD-like equation of state can substantially reduce the PBH formation threshold. The agreement with the radiation-era benchmark supports the reliability of the numerical method, while the QCD-like crossover runs show that early-universe microphysics can have an exponentially large effect on PBH abundance. This motivates future work using full three-dimensional BSSN simulations, nonspherical perturbations, and more detailed lattice-QCD-based thermodynamics.

5 Conclusion

We have studied primordial black hole formation during QCD-like equation-of-state transitions using relativistic numerical simulations. The simulation validates the standard radiation-era collapse threshold with 1.3% agreement, supporting the reliability of the numerical method. When a time-dependent QCD-like equation of state is introduced, the collapse threshold decreases substantially because the fluid temporarily loses pressure support. For $w_{\min} = 0.12$, the threshold reduction ranges from 27.0% to 44.1% depending on the crossover width.

The physical reason for this enhancement is that the QCD-like crossover affects both the background expansion and the sound speed. Since c_s^2 is not generally equal to w during a time-dependent transition, constant equation-of-state models miss part of the collapse physics. In the sign convention used here, the derivative term can either raise or suppress c_s^2 relative to w, so the key quantity is the full time history of pressure-wave propagation rather than the instantaneous value of w alone.

The resulting threshold reduction leads to an exponentially large enhancement in the predicted PBH abundance, with a peak enhancement of approximately 4.18×10^{10} for $\sigma_\delta = 0.03$. The enhancement peaks near $T \simeq 132$ MeV, corresponding to PBHs with masses of order $1M_\odot$. These results suggest that QCD-era microphysics can play an important role in PBH formation. However, they should be interpreted as a parametric sensitivity study of QCD-like softening rather than as a replacement for lattice-QCD-based numerical treatments. Future work should extend the calculation beyond spherical symmetry, use more realistic perturbation profiles, and connect the relativistic collapse calculation more directly to full lattice-QCD-based thermodynamics and three-dimensional BSSN simulations.

6 Appendix

6.1 Appendix I: Code

We are using Python to simulate basic general relativity scenarios, and PyGame to visualize our results in real-time. PyGame includes an easy-to-use interface for the simulations, and we are using the Python packages time and math to solve the basic differential equation calculations. So far, we have simulated the orbits of single particles around a Schwarzschild black hole. The particle successfully demonstrated pronounced relativistic precession around the black hole close to its Schwarzschild radius. A Github with all our raw code can be found here. In the image below, you can see that the relativistic correction causes a high amount of precession, which is more pronounced due to how close we are to the horizon (red).

6.2 Appendix II: General Relativity

In 1915, Einstein formulated the theory of general relativity, which describes gravity as the dynamical curvature of spacetime sourced by energy and momentum. The fundamental equations governing this relationship are the Einstein field equations,

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}, \tag{34}$$

where $R_{\mu\nu}$ is the Ricci curvature tensor, R is the Ricci scalar, $g_{\mu\nu}$ is the spacetime metric, Λ is the cosmological constant and $T_{\mu\nu}$ is the stress–energy tensor. The Einstein field equations form a system of coupled, nonlinear partial differential equations. Although each tensor is a 4×4 object, symmetry reduces the number of total equations to ten. These equations are generally intractable

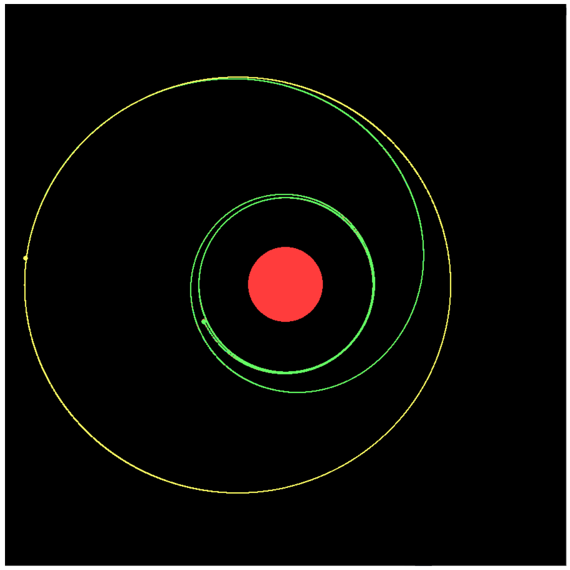

Figure 4: Image of particle motion in Newtonian (yellow) gravity and Schwarzshild (green) metric

analytically except in highly symmetric cases, motivating the use of numerical methods in studies of relativistic dynamics and gravitational collapse. The coupling constant

$$\frac{8\pi G}{c^4} \approx 2.1 \times 10^{-43}\ \mathrm{m}^{-1}\mathrm{kg}^{-1}\mathrm{s}^2 \tag{35}$$

reflects the weakness of gravity, indicating that extremely large energy densities are required to produce significant spacetime curvature.

6.3 Appendix III: Principle of Least Action

The action of a particle S is defined as follows:

$$S = \int_{t_1}^{t_2} L(x, \dot{x}, t)dt \tag{36}$$

$$\delta S = 0 \tag{37}$$

$$\frac{\delta S}{\delta x(t)} = \lim_{|\delta x(t)| \to 0} \frac{S(x(t), \dot{x}) - S(x(t) + \delta x(t), \dot{x})}{\delta x} = 0. \tag{38}$$

Recall that the Taylor expansion for a differentiable function is:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n. \tag{39}$$

The first few terms are written as:

$$f(x) = f(x_0) + \frac{f'(x_0)}{1!}(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f'''(x_0)}{3!}(x - x_0)^3 + \cdots. \tag{40}$$

Using big O notation, we will combine all the "faster than linear" terms into one for ease.

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + O(x^2). \tag{41}$$

The $O(x^2)$ just means it contains all terms of x with a power greater than or equal to 2. The Taylor expansion for a function of two variables looks something like this:

$$f(x,y) = f(x_0,y_0) + \left.\frac{\partial f}{\partial x}\right|_{x=x_0,y=y_0}(x-x_0) + \left.\frac{\partial f}{\partial y}\right|_{x=x_0,y=y_0}(y-y_0) + O(x^2,y^2). \tag{42}$$

Going back to the minimized action equations, we can rewrite the equation as:

$$\int \left(L(x,\dot{x}) + \frac{\partial L}{\partial x}\delta x + \frac{\partial L}{\partial \dot{x}}\frac{d}{dt}\delta x \right) dt. \tag{43}$$

This can be simplified using integration by parts:

$$\int \left(\frac{\partial L}{\partial x}\delta x - \delta x \frac{d}{dt}\frac{\partial L}{\partial \dot{x}} \right) dt = 0. \tag{44}$$

Since this equation must hold for any function $\delta x(t)$, we can factor out $\delta x(t)$ to obtain:

$$\int \left(\delta x \left(\frac{\partial L}{\partial x} - \frac{d}{dt}\frac{\partial L}{\partial \dot{x}} \right) \right) dt. \tag{45}$$

Since this equation always holds true, the factor next to δx must be 0; therefore:

$$\frac{\partial L}{\partial x} = \frac{d}{dt}\frac{\partial L}{\partial \dot{x}}. \tag{46}$$

This is known as the Euler-Lagrange equation.

6.4 Appendix IV: ADM Formalism

The Arnowitt–Deser–Misner (ADM) formalism is a Hamiltonian formulation of general relativity that decomposes four-dimensional spacetime into a foliation of three-dimensional spatial hypersurfaces parametrized by a time coordinate. This $3+1$ decomposition is particularly well suited for numerical implementations, as it recasts Einstein's equations into a set of constraint equations in each spatial slice and evolution equations that propagate geometry forward in time. In the ADM formalism, the spacetime line element is written as

$$ds^2 = -N^2 dt^2 + g_{ij}(dx^i + N^i dt)(dx^j + N^j dt), \tag{47}$$

where N is the lapse function, N^i is the shift vector, and g_{ij} is the induced spatial metric on each constant-time hypersurface. Greek indices (μ,ν) run over spacetime coordinates $(0,1,2,3)$, while Latin indices (i,j) run over spatial coordinates $(1,2,3)$. The lapse function N determines the rate at which proper time elapses between neighboring hypersurfaces as measured by an observer normal to the spatial slice. Physically, it encodes gravitational time dilation effects and sets the proper time separation between slices. The shift vector N^i describes how spatial coordinates shift laterally from one hypersurface to the next, accounting for the fact that observers may move relative to the normal direction. The spatial metric g_{ij} encodes the intrinsic geometry of each three-dimensional slice. This decomposition allows Einstein's field equations to be separated into constraint equations, the Hamiltonian and momentum constraints, which must be satisfied on each spatial slice, and evolution equations that determine how g_{ij} and its conjugate momentum evolve in time. In numerical relativity, these equations form the foundation for evolving relativistic spacetimes, including gravitational collapse scenarios relevant to primordial black hole formation [20].

6.5 Appendix V: BSSN Formulation

The ADM equations are conceptually natural, but in their raw form they are not usually the most stable system for long numerical evolutions. The Baumgarte–Shapiro–Shibata–Nakamura (BSSN) formulation rewrites the ADM variables in a way that improves numerical stability by separating the conformal geometry from the overall volume expansion of each spatial slice [21, 22]. In ADM, the main spatial variables are the three-metric g_{ij} and the extrinsic curvature K_{ij}. BSSN introduces a conformal decomposition

$$g_{ij} = e^{4\phi}\tilde{\gamma}_{ij}, \qquad \det(\tilde{\gamma}_{ij}) = 1, \tag{48}$$

and splits the extrinsic curvature into its trace and trace-free parts,

$$K_{ij} = e^{4\phi}\tilde{A}_{ij} + \frac{1}{3}g_{ij}K. \tag{49}$$

The evolved BSSN variables are usually taken to be

$$\left\{\phi, \tilde{\gamma}_{ij}, K, \tilde{A}_{ij}, \tilde{\Gamma}^i\right\}, \tag{50}$$

where $\tilde{\Gamma}^i = -\partial_j\tilde{\gamma}^{ij}$ are the conformal connection functions. These extra variables help convert the equations into a form that behaves better numerically. For PBH formation, BSSN is useful because collapse involves strong spacetime curvature, large gradients, and possible apparent-horizon formation. A one-dimensional Misner–Sharp calculation is efficient for spherical collapse, but a three-dimensional PBH simulation needs a formulation that can evolve nonspherical geometry without assuming exact spherical symmetry. BSSN, often combined with adaptive mesh refinement and suitable gauge choices for the lapse and shift, is one standard route to such simulations.

6.6 Appendix VI: Fluid Simulation

PBH formation is not just a problem in vacuum general relativity. The collapsing region is filled with a relativistic cosmological fluid, so the simulation must evolve both spacetime geometry and matter. In general relativity, the fluid is described by the stress–energy tensor

$$T^{\mu\nu} = (\rho + p)u^\mu u^\nu + pg^{\mu\nu}, \tag{51}$$

where ρ is the energy density, p is the pressure, and u^μ is the fluid four-velocity. Local energy-momentum conservation requires

$$\nabla_\mu T^{\mu\nu} = 0. \tag{52}$$

These equations are coupled to Einstein's field equations, so matter tells spacetime how to curve and spacetime tells the fluid how to move. In spherical symmetry, the Misner–Sharp formalism packages this coupled system into variables such as the areal radius R, the radial velocity U, the mass function M, the lapse A, and the radial metric component B. In more general numerical-relativity codes, the relativistic Euler equations are often written in conservative form so that shocks and steep gradients can be handled more reliably. The equation of state closes the system by specifying how p depends on the thermodynamic variables. In this project, the important equation-of-state input is the time-dependent softening associated with the QCD-like crossover. The sound speed is especially important for fluid evolution. It controls the characteristic speeds that enter the CFL timestep condition, and it determines how quickly pressure information travels across the overdense region.

If the timestep is too large compared with the sound-crossing time of a grid cell, the numerical solution becomes unstable. This is why the simulation uses adaptive CFL time stepping and artificial viscosity: both are practical tools for keeping the evolution stable when collapse produces sharp gradients.

6.7 Appendix VII: Einstein Toolkit

The Einstein Toolkit is a public, community-developed computational infrastructure for numerical relativity and relativistic astrophysics [23]. It is built on the Cactus framework, in which different modules, called thorns, provide different pieces of the calculation. For example, one thorn may evolve the spacetime metric, another may evolve hydrodynamic variables, another may provide mesh refinement, and another may analyze horizons or gravitational-wave output. The Einstein Toolkit is important because full numerical relativity is too large for most projects to build from scratch. A realistic three-dimensional simulation requires spacetime evolution, gauge choices, constraint monitoring, mesh refinement, boundary conditions, hydrodynamics, and output analysis. The Toolkit packages many of these ingredients into a standard infrastructure. It has been used for black holes, neutron stars, gravitational collapse, and cosmological spacetimes. For future PBH work, the Einstein Toolkit would be useful as a route from the spherical Misner–Sharp model used in this paper to more general simulations. In particular, a three-dimensional calculation could test whether the threshold reductions found here survive when the perturbation is nonspherical, when the profile shape is varied, or when more realistic spatial structure is included. The tradeoff is complexity: a full Einstein Toolkit calculation requires substantially more setup, validation, and computational resources than a one-dimensional spherical solver.

6.8 Appendix VIII: GRChombo

GRChombo is another numerical-relativity code designed for adaptable simulations of Einstein's equations, especially in problems where adaptive mesh refinement is important [24]. It is built on the Chombo adaptive mesh refinement library and is written for high-performance simulations of strong-field gravitational systems. GRChombo has been used in a range of numerical-relativity and early-universe applications, including scalar-field collapse, black-hole dynamics, and beyond-standard-model cosmological scenarios. For PBH formation, the value of GRChombo is that collapse is localized. Most of the computational domain may remain close to a smooth expanding background, while a small overdense region develops large gradients and strong curvature. Adaptive mesh refinement is designed for exactly this kind of problem: the grid can be made fine where collapse is occurring and coarser where the solution is smoother. This makes three-dimensional collapse simulations more computationally feasible than using a uniformly fine grid everywhere. A GRChombo extension of this project would replace the spherical Misner–Sharp restriction with a full three-dimensional spacetime evolution, likely using BSSN or a closely related formulation. That would make it possible to test nonspherical perturbations, angular structure, and profile-dependent effects. Such a calculation would also allow a more direct comparison with recent full-GR studies of PBH formation in other mechanisms, such as collapsing domain walls [19]. The present paper should therefore be viewed as a lower-dimensional, controlled step toward that broader numerical-relativity program.

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